Authentication Functions Overview

Authentication requirement – Authentication function – MAC – Hashfunction – Security of hash function and MAC – SHA –Digital signature and authentication protocols – DSS- Entity Authentication: Biometrics, Passwords, Challenge Response protocols- Authentication applications –Kerberos, X.509

Message Authentication and Integrity

1. Authentication Requirement

Authentication means verifying that the message comes from the genuine sender and has not been altered.

Main Requirements

1. **Message Integrity**
   Ensures message is not modified during transmission.
2. **Authentication**
   Confirms identity of sender.
3. **Non-repudiation**
   Sender cannot deny sending the message.
4. **Freshness**
   Prevents replay attacks using old messages.
5. **Confidentiality (optional)**
   Protects message content from unauthorized access.

---

2. Authentication Function

An authentication function verifies the originality and integrity of a message.

Types of Authentication Functions

(a) Message Encryption

Entire message is encrypted using a secret key.

- Provides confidentiality and authentication.
- Receiver decrypts and verifies sender.

(b) Message Authentication Code (MAC)

A fixed-size code generated using:

- message
- secret key

Receiver recalculates MAC and compares it.

(c) Hash Function

Produces fixed-length digest from message.

- Faster than encryption
- Used in digital signatures and integrity checking

---

3. MAC (Message Authentication Code)

MAC is a cryptographic checksum used for authentication and integrity.

Working

$$MAC = C(K, M)$$

Where:

- K = Secret key
- M = Message

Sender sends:

- Message
- MAC value

Receiver:

- Uses same key
- Generates MAC again
- Compares values

If same → message is authentic.

Features of MAC

1. Detects message modification

2. Authenticates sender
3. Uses secret key
4. Faster than digital signature

Types of MAC

1. HMAC
2. CBC-MAC
3. CMAC

4. Hash Function

A hash function converts variable-length input into fixed-length output called hash value or message digest.

$$H(M) = Digest$$

Properties of Good Hash Function

1. Fixed output length
2. Fast computation
3. One-way function
4. Collision resistant
5. Sensitive to small changes

Applications

- Password storage
- Digital signature
- Data integrity
- Blockchain
- File verification

5. Security of Hash Function and MAC

Security Requirements of Hash Function

(a) Preimage Resistance

Given hash h, difficult to find message m.

(b) Second Preimage Resistance

Difficult to find another message with same hash.

(c) Collision Resistance

Hard to find two messages producing same hash.

Security of MAC

MAC should resist:

1. Forgery attacks
2. Replay attacks
3. Brute-force attacks
4. Key guessing attacks

Strong secret keys improve security.

---

6. SHA (Secure Hash Algorithm)

SHA is a family of cryptographic hash algorithms designed by NSA.

SHA Family

1. SHA-1
2. SHA-2
3. SHA-3

SHA-1

- Produces 160-bit hash
- Now considered weak

SHA-2

Includes:

- SHA-224
- SHA-256
- SHA-384
- SHA-512

Widely used today.

SHA-3

- Latest version
- Based on Keccak algorithm
- More secure against future attacks

Applications of SHA

1. Password hashing
2. SSL/TLS
3. Digital signatures
4. Blockchain
5. Integrity verification

---

7. Digital Signature

Digital signature provides:

1. Authentication
2. Integrity
3. Non-repudiation

Uses asymmetric cryptography.

Working of Digital Signature

Sender Side

1. Create message digest using hash function
2. Encrypt digest using sender's private key

3. Send message + signature

Receiver Side

1. Decrypt signature using sender's public key
2. Generate digest again
3. Compare both digests

If equal → valid signature.

Advantages

1. Prevents forgery
2. Provides legal authenticity
3. Ensures integrity

---

8. Authentication Protocols

Protocols used to verify communicating entities.

Types

1. Password-based authentication
2. Challenge-response authentication
3. Token authentication
4. Biometric authentication

---

9. DSS (Digital Signature Standard)

DSS is a U.S. standard for digital signatures.

Based on:

- DSA (Digital Signature Algorithm)

Uses:

1. SHA for hashing
2. Public key cryptography

Features

1. Authentication
2. Integrity
3. Non-repudiation

---

10. Entity Authentication

Entity authentication verifies identity of user/device.

(a) Biometrics

Authentication using physical or behavioral characteristics.

Examples

1. Fingerprint
2. Iris scan
3. Face recognition
4. Voice recognition

Advantages

- Difficult to duplicate
- Convenient

Disadvantages

- Expensive
- Privacy concerns

(b) Passwords

Most common authentication method.

Types

1. Static password
2. One-time password (OTP)

Weaknesses

- Guessing
- Dictionary attacks

- Phishing

(c) Challenge-Response Protocol

Used to prevent replay attacks.

Working

1. Server sends random challenge
2. User encrypts/signs challenge
3. Server verifies response

No password sent directly.

---

11. Authentication Applications

Applications where authentication is essential:

1. Online banking
2. E-commerce
3. Secure email
4. VPN
5. Cloud security

---

12. Kerberos

Kerberos is a network authentication protocol using secret-key cryptography.

Developed at:
Massachusetts Institute of Technology

Components

1. Client
2. Authentication Server (AS)
3. Ticket Granting Server (TGS)
4. Service Server

Working of Kerberos

1. User logs in
2. AS provides Ticket Granting Ticket (TGT)
3. TGS issues service ticket
4. User accesses service securely

Advantages

1. Mutual authentication
2. Single sign-on
3. Secure against eavesdropping

Disadvantages

1. Time synchronization required
2. Central server dependency

---

13. X.509

X.509 is a standard for public key certificates.

Used in:

- SSL/TLS
- HTTPS
- Digital certificates

X.509 Certificate Contains

1. Version
2. Serial number
3. Issuer name
4. Subject name
5. Public key
6. Validity period
7. Digital signature

Uses

1. Secure web communication
2. Public key infrastructure (PKI)
3. Authentication in networks